

PalmSource®

Conference and Expo



The Palm OS® Roadmap

Steve Sakoman, Chief Product Officer



Roadmap: "A Plan to Guide Progress to a Goal"

What you need to know to guide your planning and development efforts over the next two years



Our Goal

The Journey

The First Step

The Take-Away

Our Goal for Palm OS® Is Simple

To enhance the way people worldwide

- Communicate
- Organize their lives
- Stay informed
- Conduct business
- Utilize downtime





Our Design Philosophy

We remain true to our roots

- Practical innovation
- Intuitive simplicity
- Trustworthy technology

One Size Does Not Fit All…

A variety of Palm OS® devices beyond PDAs





Our Goal

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An Evolutionary Path to the Future

The focus of the first step…

- ARM Support
  - Performance
  - Flexibility
  - Choice
- Compatibility
  - Smooth transition
  - Software investment protection for users
  - Market protection for developers

An Evolutionary Path to the Future

Followed by a strong focus on capability







An Evolutionary Path to the Future

Security needs

- User Data
- Network connections
- Software installation



An Evolutionary Path to the Future

Security features

- Extensible, Pluggable framework
- Encryption/Decryption
- Authentication/Authorization
- Secure Network Connections
- Code Signature Generation/Verification
- Certificate Management



An Evolutionary Path to the Future

Communication needs

- Voice
- Email
- Messaging
- Synchronization
- Web access



An Evolutionary Path to the Future

Communication features

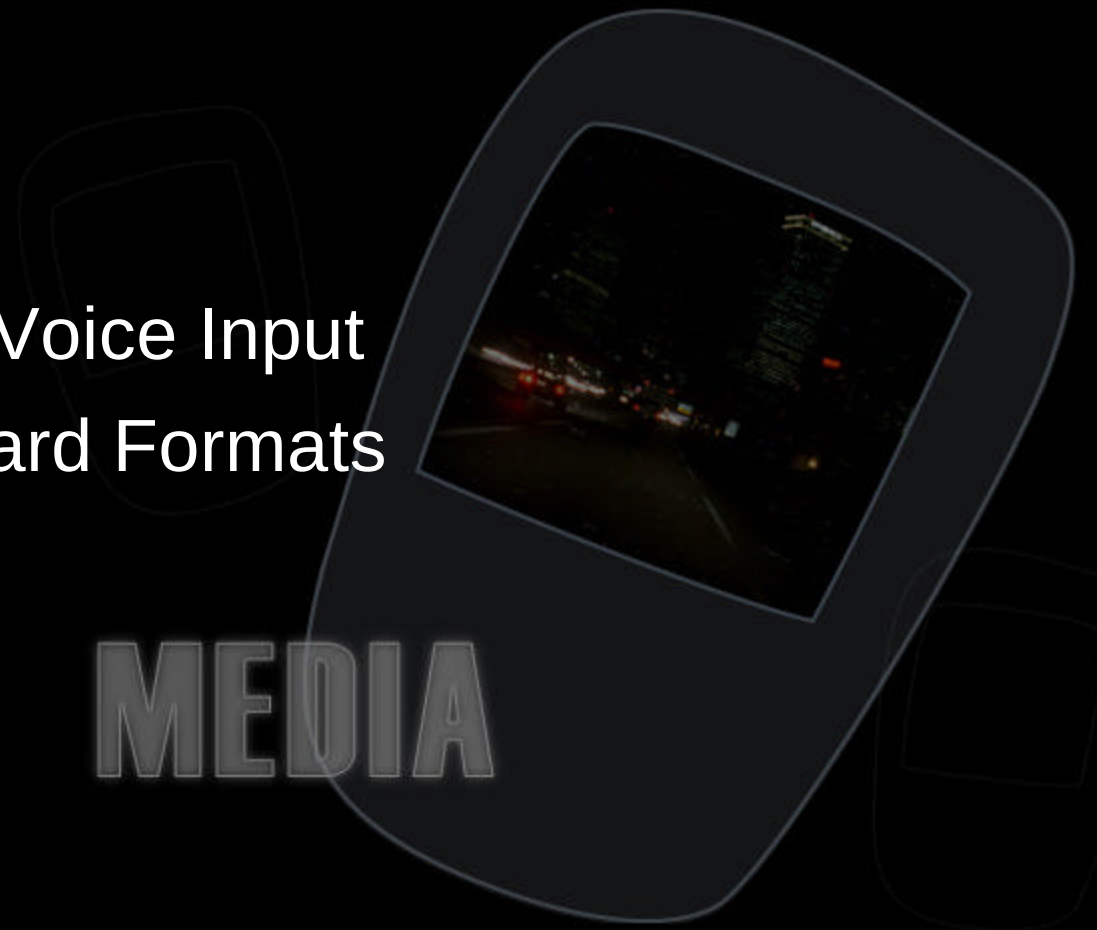
- Extensible, Pluggable Framework
- Unified IO Subsystem
- Telephony
- Wireless WAN/LAN
- Web technologies
- Proxy-less browser



An Evolutionary Path to the Future

Media needs

- Sound
- Video
- Stylus, Keyboard, Voice Input
- Playback of Standard Formats



An Evolutionary Path to the Future

Media features

- Extensible, Pluggable Framework
- Screen Resolutions
- Input Methods
- UI Themes
- Audio/Video File Formats/Codecs





Agenda

- Our Goal
- The Journey
- The First Step
- The Take-Away



The First Step: Palm OS® Version 5

A foundation for the future—Performance

- ARM 7 architecture and above
- Support for Intel, Motorola, and TI CPUs
- Fully Native 32-bit OS
- Builds upon Palm OS 4.1 APIs



The First Step: Palm OS 5

A foundation for the future—Compatibility

- Emulation of 68K application binaries
- Native ARM execution of Palm OS calls
- Emulation performance > 68K native
- Power consumption = 68K



The First Step: Palm OS 5

A foundation for the future—Security

- New Crypto Manager for data protection
- SSL for transactions, email, web browsing
- Licensee options for VPN and SecureID



The First Step: Palm OS 5

A foundation for the future—Communications

- 802.11b Wireless Networking
  - Documented low-level TCP APIs
  - Enables 3rd-party driver development



The First Step: Palm OS 5

A foundation for the future—Media

- Sound Manager APIs
  - Digital audio output and recording (mono/stereo)
  - 16-channel mixer with per channel level control
  - Licensee option for hardware acceleration



The First Step: Palm OS 5

A foundation for the future—Media

- High Density Display APIs
  - Backward compatible
  - Legacy applications benefit from new double density fonts and UI widgets
  - Double density icons, bitmaps
  - Double density Latin and Japanese fonts
  - Licensee option for resolutions other than 320x320 and 160x160



The First Step: Palm OS 5

A foundation for the future—Media

- User Interface Color Themes
  - User selectable via General Preferences Panel
  - Licensee opportunity for custom themes



The First Step: Palm OS 5

A foundation for the future—Tools

- CDK update for Visual Studio.net and Mac OS X
- Palm Universal Debugger with serial support
- Constructor and PalmRez
  (68K resource support)



Our Goal

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20 Million Palm OS Devices

We have a vision for the future

The first step is here for you to work with *now*:

Palm OS® Version 5



An evolutionary approach protects:

Your work

Your knowledge

Your investments

Palm OS 5



Seeding programs give early access

Pick up your CD today

Head to the labs to try your app!

Palm OS 5

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Understanding the New Technologies

David Fedor, Architect/Evangelist



All the new technologies
of PalmSource® 2002
in one session. (Brace yourself.)



The Big Picture

- Why the ARM chip
- What switching chips means
- Compatibility of 68k applications
- Impact to your software



The Big Picture: Palm OS® Version 5

- Fully native OS
- New Kernel and low-level software
- Chip independent
- Fully 32-bit (not as if that means much to users)



How OS Releases Are Done

- Core OS work: same for all hardware
- Modular OS components and applications
- Palm OS Ready program, and the DAL
- Licensee customization and extensions
- When devices will arrive, and their specifics



Hardware demos!



Development Tools

- Your existing ones: CodeWarrior, gcc/prc-tools, higher level tools, etc.
- A new one: Palm OS Simulator
  - Keep using Palm OS Emulator with 4.x ROMs
- Native ARM coding

- Come to session 100 for details



Compatibility

- Common bugs
- New and old Accessors
- New Notifications
- Working well with all Palm OS devices



Compatibility

- Test on the Palm OS 5 Simulator
- Test on Pose (4.x Debug ROMs)
- Test on ARM-based Hardware
- … Before your customers do!



Compatibility

- Pose's warnings over the years were because of what's in Palm OS 5
- Palm OS is moving forward quickly and yet maintains compatibility
  - Just can't assume 1.0's implementations
- Come to session 103 for many more details



New Core Palm OS additions

- Security APIs
  - 504: Security Roadmap (and feedback)
  - 111: Security APIs for Palm OS 5 (Cert. Mgr, crypto)
  - 505: Securing Your Internet/intranet (VPN, SSL)
- High Density Screen API
  - Session 112
- Sampled Sound
  - Session 112
- 802.11b driver/framework
- UI Color schemes



Other Crucial Talks About New Technology

- Bluetooth
  - Session 301
- New Desktop APIs
  - Sessions 101, 102
- Targeting All Devices
  - Session 103



Next Big OS Release

- Tons of new APIs
- Full native ARM applications
  - Thus, new toolchain
  - Endianness, wire formats, etc.
  - New application environment, event handling



Next Steps

- Getting the Palm OS 5 Simulator
- Getting the 4.0 SDK update
- Hardware in the labs
- Engineers in the labs
  - (Look for the ribbons)



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